ANALYTICAL CUMULATIVE EXAM

April 20, 1996

This cume tests your understanding of the method of internal standards. Internal standards are desirable whenever unknown losses of sample are likely to occur during handling or analysis. If an known quantity of standard (different from the analyte) is added to the unknown prior to any handling losses, the ratio of standard and analyte will remain constant thereafter, because the same fraction of each is lost in any operation. The known standard added to the analyte is the internal standard.

The method of internal standard is very effective in capillary gas chromatography where the amount of sample that actually enters the column is uncertain. Consider a mixture of unknown analyte X, and standard S, which are injected into a capillary GC. Each species will give a resulting peak area A_x and A_s, and we do know for the range of concentrations of both S and X the peak area is directly proportional to the injected concentration of each, or

$$A_X = k_X[X] \text{ and } A_S = k_S[S]$$

Where k_X and k_S are the response factors the GC detector has for each compound.

A GC analysis using an internal standard is done with two separate injections. One injection contains a known mixture of analyte and internal standard, $[X]_{\text{known inject}}$, $[S]_{\text{known inject}}$. The second injection contains a mixture of unknown analyte with a known concentration of internal standard $[X]_{\text{unknown inject}}$, $[S]_{\text{unknown inject}}$.

1. If the area of the peaks is proportional to the injected concentration through the sensitivity factors listed above, derive an equation in the following form which shows how ratios can be used to not only compensate for sample losses, but also for the sensitivity factors as well.

$$\left(\frac{\textit{concentration ratio of unknown injection}}{\textit{concentration ratio of known injection}} \right) = \left(\frac{\textit{area ratio unknown injection}}{\textit{area ratio known injection}} \right)$$

2. In the experiment 10 mL of 0.084M X is mixed with 10 mL of 0.067M S and diluted to 25 ml. When injected the resulting areas of the internal standard and the analyte are: A_S = 375 and A_X = 425. (Areas are measured in arbitrary units). To analyze the unknown, 10 mL of 0.15M S is mixed with 5mL of unknown X solution and diluted to 25 mL. The resulting areas are: A_S = 554 and A_X = 565. Find the concentration of X in the 10 mL unknown sample.